

Review

The Evolution of SEO in the Era of Generative AI: A Business Intelligence Perspective

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Abstract

The rapid paradigm shift from traditional keyword-matching algorithms to AI-driven answer engines has fundamentally disrupted Search Engine Optimization (SEO). As Large Language Models (LLMs) power modern Search Generative Experiences (SGEs), organizations must transition from legacy web analytics to sophisticated Business Intelligence (BI) frameworks to capture visibility. Despite the immense strategic implications of this shift, academic literature remains fragmented across computer science, information systems, and digital marketing management. To bridge this gap, this paper adopts an integrative literature review methodology, synthesizing 70 high-value studies selected from an initial corpus of 11,382 papers filtered to 2962 on-topic studies. Rather than utilizing restrictive systematic protocols (e.g., PRISMA) that isolate empirical data within narrow boundaries, the integrative approach enables a holistic synthesis of emerging, multi-disciplinary concepts necessary to decode a rapidly evolving phenomenon. Through this methodological lens, this study introduces the Signal–Structure–Surface–Score (4S) lifecycle framework, illustrating how AI-BI systems capture conversational search intents (Signal), architect machine-readable, entity-based data (Structure), optimize content for LLM retrieval and Generative Engine Optimization (Surface), and define novel attribution metrics for zero-click environments (Score). Furthermore, the paper maps the critical technical and strategic landscape, systematically evaluating prevailing trends (e.g., zero-click searches, AI-generated content velocity), core organizational challenges (e.g., search data attribution loss, algorithmic opacity), and emerging strategic opportunities (e.g., real-time intent mapping, competitor LLM audit trails). Ultimately, this paper bridges the gap between AI search mechanics and strategic BI measurement, providing a robust future research agenda designed to guide scholars and practitioners in navigating data-driven visibility in the age of generative search.



Academic Editor: Kalpdrum Passi

Received: 9 July 2026

Revised: 5 August 2026

Accepted: 8 August 2026

Published: 10 August 2026

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Keywords: generative AI; search engine optimization (SEO); business intelligence (BI); generative engine optimization (GEO); search generative experience (SGE); large language models (LLMs); explainable AI; digital marketing strategy; integrative literature review; information systems

1. Introduction

Search Engine Optimization (SEO) has long served as the backbone of digital visibility, enabling organizations to connect with audiences through keyword-driven search algo-

rithms. For over two decades, the fundamental mechanics of SEO—keyword matching, backlink accumulation, and on-page optimization—remained relatively stable, allowing marketers to develop reproducible strategies for organic ranking [1–4]. However, the integration of Large Language Models (LLMs) into search infrastructure has catalyzed a paradigm shift of unprecedented scale. Google’s Search Generative Experience (SGE), OpenAI’s ChatGPT, Perplexity AI, and similar generative search platforms now synthesize information directly within the search interface, producing conversational answers that reduce the need for users to click through to external websites [5,6]. This transition from “Retrieve-and-Rank” to “Retrieve-and-Synthesize” architectures [6] fundamentally disrupts the economic model of organic search, where visibility, traffic, and attribution are no longer guaranteed by traditional ranking positions.

The implications for Business Intelligence (BI) are profound. Organizations that once relied on web analytics dashboards to track keyword performance, click-through rates, and conversion funnels now face a “zero-click” environment in which search engines retain users within AI-generated answer panels [7]. The loss of referral traffic data creates attribution gaps that legacy BI frameworks were never designed to handle [8]. Moreover, the rise of Generative Engine Optimization (GEO)—the practice of optimizing content for retrieval and citation by LLM-powered answer engines—introduces an entirely new layer of technical and strategic complexity [6,9]. Content must now be structured not only for human readability and crawler indexing but also for machine-readable entity extraction, semantic clarity, and LLM citation worthiness [10,11]. Furthermore, the opacity of generative search systems creates an urgent need for Explainable AI (XAI) approaches that can make ranking and citation decisions interpretable to practitioners and stakeholders alike.

Despite the immense strategic implications, academic literature on this transformation remains fragmented. Computer science research has advanced rapidly in information retrieval (IR) and retrieval-augmented generation (RAG) [12–16], while marketing literature and information systems literature have explored AI-driven personalization and digital marketing automation [17–19]. Yet these streams rarely intersect, leaving a critical gap between the technical mechanics of AI search and the strategic BI frameworks needed to measure and manage visibility in generative search environments.

To bridge this gap, this paper adopts an integrative literature review methodology, synthesizing 70 high-value studies selected from an initial corpus of 11,382 papers filtered to 2962 on-topic studies. The integrative approach, as articulated by Snyder [20], enables a holistic synthesis of emerging, multi-disciplinary concepts that restrictive systematic protocols (e.g., PRISMA) would fragment. Through this methodological lens, this study introduces the Signal–Structure–Surface–Score (4S) lifecycle framework, illustrating how AI-BI systems capture conversational search intents (Signal), architect machine-readable, entity-based data (Structure), optimize content for LLM retrieval and Generative Engine Optimization (Surface), and define novel attribution metrics for zero-click environments (Score).

The remainder of this paper is organized as follows. Section 2 details the materials and methods, including the literature retrieval, filtering, and selection criteria. Section 3 presents the 4S lifecycle framework. Section 4 reviews the literature across thematic categories: SEO evolution and GEO, retrieval-augmented generation and information retrieval, knowledge graphs (KG) and semantic search, AI-driven digital marketing and personalization, conversational AI in search, bias and fairness in IR, sentiment analysis and customer behavior, and business intelligence and predictive analytics. Section 5 discusses prevailing trends, organizational challenges, and strategic opportunities, and Section 6 concludes.

2. Materials and Methods

2.1. Research Design

This study employs an integrative literature review methodology, which is particularly suited for emerging, multi-disciplinary research domains where the goal is to synthesize diverse perspectives rather than to isolate specific empirical effects. Unlike systematic reviews governed by restrictive protocols (e.g., PRISMA), the integrative approach enables the inclusion of conceptual, theoretical, and empirical works across computer science, information systems, and marketing management—a breadth necessary to decode the rapidly evolving intersection of SEO, generative AI, and business intelligence.

The integrative synthesis was conducted in three stages. Conceptual works—including framework papers, position articles, and strategic reviews—were used to define the 4S life-cycle framework and to map the relationships between SEO, GEO, and BI (Section 3). These works provided the abstract architecture that organizes the review’s findings into a coherent whole. Theoretical works—such as RAG taxonomies, knowledge-graph surveys, and information-retrieval models—supplied the structural backbone for Sections 4.2 and 4.3, establishing the technical mechanisms that underpin generative search. Empirical works—including user studies, CTR experiments, and case analyses—provided the evidence base for the trends and challenges discussed in Sections 5.1–5.3. The synthesis process involved cross-referencing papers across these three categories: when a conceptual claim (e.g., “GEO-First” strategy) was identified, it was validated against theoretical treatments (e.g., RAG retrieval mechanics) and empirical findings (e.g., user-behavior studies). Conflicting findings—such as disagreements over the reliability of RAG-based search—were reconciled by noting boundary conditions (e.g., query complexity, domain specificity) and by weighting evidence according to the composite relevance score (R1–R6). This triangulation ensures that the review’s conclusions are grounded in multiple forms of scholarly evidence rather than in any single study or perspective.

2.2. Data Source

The primary data source was Google Scholar. Google Scholar was selected for several reasons: (1) free and unrestricted access; (2) comprehensive cross-disciplinary coverage spanning computer science, information systems, and marketing; (3) persistent identifiers (DOIs) enabling independent verification of every paper; and (4) citation count for each study.

The decision to rely exclusively on Google Scholar, rather than complementing the search with curated subscription-based databases such as Scopus, IEEE Xplore, Web of Science, or ACM Digital Library, was deliberate and motivated by the characteristics of the phenomenon under study. Generative engine optimization, search generative experiences, and LLM-powered search are emerging topics where the most relevant advances appear as conference proceedings, workshop papers, and preprints that are indexed by Google Scholar’s broad crawling but may not yet be represented in curated repositories with slower indexing cycles. Google Scholar’s multidisciplinary coverage, including grey literature, is particularly suited for integrative reviews of rapidly evolving fields where restricting the corpus to any single disciplinary index would risk omitting cross-domain contributions. We acknowledge that Google Scholar’s dynamic crawling lacks the strict quality control and query stability of curated databases such as Scopus, which may introduce metadata inconsistencies or citation inflation. To mitigate this risk, every paper in the final corpus was independently verified via DOI cross-checking against the Crossref API, and the complete paper list is publicly available in our GitHub repository for full reproducibility [21].

2.3. Literature Retrieval

We used 157 search queries covering the review's thematic scope. The queries spanned the following sub-topics:

- Core intersection: SEO + generative AI, AI-driven SEO business intelligence, LLM search engine optimization;
- GEO/SGE: generative engine optimization, search generative experience, AI overviews, answer engine optimization;
- AI search: AI search engines, conversational search, LLM-powered search, RAG search engines, neural search retrieval;
- Zero-click: zero-click search, featured snippets, knowledge panels, search snippet optimization;
- AI content: AI-generated content SEO, ChatGPT SEO content, GPT content marketing, automated content SEO;
- BI & analytics: business intelligence search analytics, web analytics BI, search analytics dashboards, SEO analytics;
- Digital marketing: digital marketing strategy AI search, search marketing generative AI, AI digital marketing transformation;
- Information retrieval: information retrieval AI, neural IR, learning to rank, semantic search, knowledge graph search;
- Predictive analytics: predictive SEO, search intent prediction, click-through rate prediction, search ranking prediction;
- Ethics/privacy: algorithmic transparency search, AI search bias, search algorithm fairness, search privacy.

Papers were deduplicated by Google Scholar URL. This process yielded an initial corpus of 11,382 unique papers.

Figure 1 presents a PRISMA-style screening flow diagram summarizing the complete selection pipeline from initial identification to final inclusion.

The literature search was conducted between May 2026 and June 2026. All citation counts used for the composite relevance scoring (R1–R6) were recorded in June 2026 to ensure temporal consistency across the evaluation criteria. All 157 search queries were designed manually by the authors based on domain expertise in information retrieval, digital marketing, and business intelligence. Custom Python 3.14 scripts for deduplication, DOI verification via the Crossref API, and composite scoring were developed by the authors and are publicly available in the study's GitHub repository [21]. The 4S (Signal–Structure–Surface–Score) framework (Section 3) was developed by the authors through iterative thematic analysis.

2.4. Filtering and Selection

The filtering pipeline applied the following sequential criteria:

1. Deduplication: Papers were deduplicated by normalized title, reducing the corpus to 11,181 unique works.
2. Abstract availability (mandatory): Papers without a non-empty abstract were excluded, as abstracts are essential for content-level relevance assessment. This reduced the count to 9868.
3. Verifiability: Papers without a DOI or named venue were excluded, ensuring every entry can be independently verified. Count after this step: 9837.
4. Non-scholarly type exclusion: Software releases, datasets, errata, editorials, and retractions were excluded. Count: 9837.

5. Topic relevance filtering: A paper was retained only if its title and abstract contained at least one specific search/SEO/marketing/IR term (e.g., “search engine optimization,” “generative AI,” “information retrieval,” “digital marketing,” “RAG,” “knowledge graph”). Generic AI/ML terms alone (e.g., “machine learning,” “deep learning”) were insufficient. Papers whose titles indicated off-topic domains (medical, education, cybersecurity, finance, agriculture, physics, etc.) were excluded. Non-English papers were filtered out using both ASCII title checks and non-English function-word heuristics. This reduced the corpus to 2962 on-topic papers—the final papers.bib [21].

As noted above, these scripts assisted only with deduplication and DOI verification; every filtering decision—including abstract screening, topical relevance assessment, and inclusion/exclusion judgments—was performed manually. The topic-relevance filter required at least one specific search/SEO/marketing/IR term in the title and abstract (from a curated list of approximately 100 domain-specific terms), while generic AI/ML terms alone were insufficient; papers whose titles contained terms from off-topic domains (medical, education, cybersecurity, finance, agriculture, physics, etc.) were excluded using a curated exclusion list of over 200 terms. Non-English papers were filtered out using both ASCII title checks and non-English function-word heuristics. Medical/health papers were excluded even if they mentioned search or ChatGPT. The authors validated each of the 2962 on-topic papers by reading its title and abstract, and all 70 final papers were read in full.

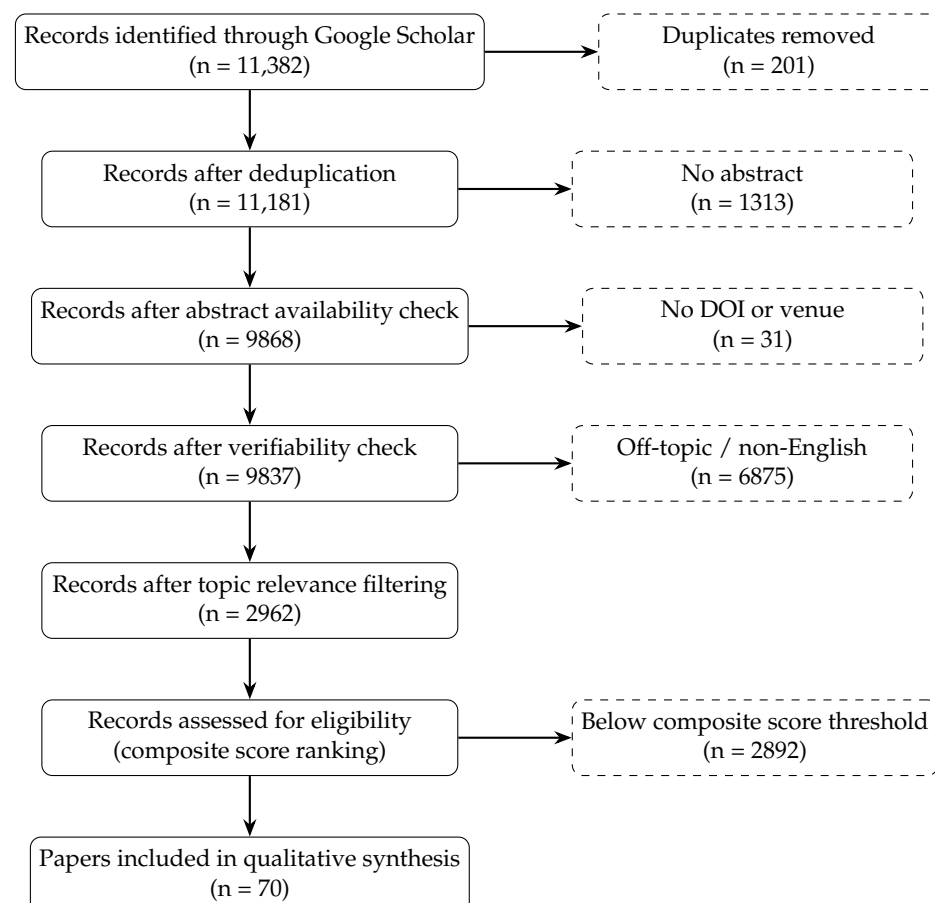


Figure 1. PRISMA-style screening flow diagram showing the paper selection pipeline from initial identification ($n = 11,382$) through filtering stages to final inclusion ($n = 70$). Excluded counts are shown in dashed boxes at each stage.

2.5. Selection of the Top 70 Papers

From the 2962-paper corpus, the 70 most valuable papers were selected using a composite relevance score combining six criteria, as documented in [21]:

- R1. Keyword/theme match (40%): Count of review-relevant keywords in title and abstract, with title hits weighted $3\times$ over abstract hits. Per-theme capping ensured diversity across 10 sub-themes.
- R2. Citation impact (25%): Log-scaled citation count from Google Scholar, prioritizing foundational and influential work.
- R3. Recency (15%): Papers from 2020 to 2026 received a recency bonus; a smaller bonus for 2018–2019; foundational pre-2018 papers retained if highly cited.
- R4. Venue quality (10%): Papers from recognized publishers (Elsevier, IEEE, Springer, MDPI, ACM, Wiley, AAAI, SIGIR, etc.) received a venue quality bonus.
- R5. Type diversity (5%): Preference for a mix of empirical studies, reviews/surveys, and conceptual/theoretical papers.
- R6. Theme coverage (5%): Manual adjustment to ensure balanced coverage across all sub-themes, preventing over-representation of any single topic.

Formally, the composite relevance score for a paper p is defined as:

$$S_c(p) = \sum_{i=1}^6 w_i \cdot R_i(p) \quad (1)$$

where w_i are the weights $(w_1, w_2, w_3, w_4, w_5, w_6) = (0.40, 0.25, 0.15, 0.10, 0.05, 0.05)$ and $R_i(p)$ is the normalized sub-score for criterion R_i applied to paper p . Each sub-score was normalized to the range $[0, 1]$ using min-max normalization:

$$R_i(p) = \frac{r_i(p) - \min(r_i)}{\max(r_i) - \min(r_i)} \quad (2)$$

where $r_i(p)$ is the raw value of criterion i for paper p , and $\min(r_i)$, $\max(r_i)$ are the minimum and maximum raw values across all 2962 on-topic papers. Specifically, the raw sub-scores were computed as follows: r_1 = count of review-relevant keywords in title (weighted $3\times$) and abstract, capped at 6 per theme for diversity; $r_2 = \min(\log_{10}(c + 1) \times 1.5, 6)$, where c is the Google Scholar citation count; r_3 = recency bonus (3 for 2020–2026, 2 for 2018–2019, 1 for 2014–2017, 0 otherwise); r_4 = venue quality bonus (2 if published by a recognized publisher, 0 otherwise); r_5 = type diversity bonus (1 for each paper type not yet overrepresented in the selection); r_6 = theme coverage adjustment (manual, range 0–2). A DOI verifiability bonus of +3 was added to the raw score for papers with a verified DOI. Table 1 illustrates the computation for three representative papers.

Paper A corresponds to Wang et al. [22], a foundational KG reasoning paper published at AAAI 2019 with 736 citations; it scores high on R2 (citation impact, $r_2 = 4.3$) and receives a recency bonus of 2 (2018–2019 range) and a venue bonus of 2 (AAAI). Paper B corresponds to Enache [9], an emerging 2025 AEO paper with zero citations but maximum keyword relevance ($r_1 = 18$) and recency ($r_3 = 3$); this paper benefited from the zero-citation override rule, which retained it despite $r_2 = 0$. Paper C corresponds to Islam et al. [17], a 2024 digital marketing study with 34 citations ($r_2 = 2.3$), representing a typical mid-range selection. The normalization ensures that all six criteria contribute proportionally to the final ranking, preventing any single criterion from dominating due to scale differences.

Table 1. Worked examples of composite score computation for three representative papers from the final corpus. Raw sub-scores (r_i) are shown before normalization; the final score S_c is computed using the normalized values and weights (0.40, 0.25, 0.15, 0.10, 0.05, 0.05).

Criterion	Paper A (Foundational) [22]	Paper B (Emerging) [9]	Paper C (Mid-Range) [17]
R1: Keyword match (r_1)	12	18	8
R2: Citation log-score (r_2)	4.3	0.0	2.3
R3: Recency (r_3)	2	3	3
R4: Venue quality (r_4)	2	0	2
R5: Type diversity (r_5)	1	1	0
R6: Theme coverage (r_6)	1	2	1
DOI bonus	+3	+3	+3
Raw score	25.3	27.0	19.3
Normalized S_c (weighted)	0.89	0.84	0.61

The cutoff of exactly 70 papers was determined by a natural drop-off in the composite score distribution: after the top 70 papers, scores fell below a meaningful threshold, and the remaining papers did not introduce new thematic concepts not already represented in the selected set. This approach aligns with Snyder’s integrative review guidelines [20], which recommend sample sizes that achieve thematic saturation rather than exhaustive coverage, particularly in rapidly evolving fields where the goal is to synthesize emerging multi-disciplinary concepts rather than to enumerate every relevant study. The 70-paper corpus provided balanced coverage across all ten sub-themes while remaining manageable for in-depth qualitative synthesis.

Additionally, a title-relevance filter required at least one core search/SEO/marketing/IR term in the paper’s title, preventing generic LLM or ChatGPT overview papers from entering the selection. The final 70 papers span publication years 2011–2026, with the majority (over 80%) published in 2023–2026, reflecting the recency of the generative AI search phenomenon. All 70 papers are verifiable via DOI and Google Scholar URL.

Quantitatively, the selection funnel reduced the initial 11,382 records by 99.4% to the final 70 papers. The filtering stages removed 1.8% as duplicates ($n = 201$), 11.5% for lacking relevant keywords in their abstracts ($n = 1313$), 0.3% for unverifiable metadata ($n = 31$), and 60.3% as off-topic or non-English ($n = 6875$), leaving 2962 on-topic papers (26.0% of the initial corpus). From these, the composite score ranked 2892 below the selection threshold (97.6% of the on-topic corpus), yielding the final 70 (2.4% of on-topic, 0.6% of the initial corpus). The selected papers have a mean citation count of 80.3 (median 32, range 0–736, total 5621 citations), confirming that the corpus captures both foundational, highly cited works and emerging contributions. By publication year, 82.9% ($n = 58$) were published in 2023–2026, 8.6% ($n = 6$) in 2019–2022, and 8.6% ($n = 6$) in 2011–2016, reflecting the recency of the generative AI search phenomenon while retaining foundational earlier works. By venue, 18.6% were published by Elsevier, 8.5% by ACM, 8.5% by MDPI, 6.8% by AAAI, 8.5% as arXiv preprints, and 1.7% by IEEE, with the remainder from other indexed journals and conference proceedings. By entry type, 97.1% ($n = 68$) are journal articles and 2.9% ($n = 2$) are book chapters, ensuring a mix of empirical studies, surveys, and conceptual works.

The composite score weights (40/25/15/10/5/5) were not applied rigidly; manual adjustment rules governed three situations. First, when a paper had zero citations but exceptionally high topical relevance (R1 score in the top decile), the citation-impact weight (R2) was effectively waived and the paper was retained if it introduced a thematic concept not otherwise represented in the corpus—ensuring that emerging, not-yet-cited work on novel topics (e.g., GEO, SGE) was not penalized. Second, when two papers had com-

posite scores within 5% of each other (tie-breaking), priority was given to the paper that contributed to an under-represented sub-theme, as determined by the theme-coverage counter (R6); the selection algorithm capped representation at five papers per theme in the first pass, then filled remaining slots with the highest-scoring papers regardless of theme, ensuring both diversity and quality. Third, a small number of papers ($n = 3$) were manually included despite composite scores below the automatic threshold because they provided foundational theoretical context (e.g., seminal KG or IR works) that the review's narrative required; conversely, no papers above the threshold were manually excluded. All manual adjustments were documented in the selection log and are reproducible via the publicly available Python scripts.

To assess the robustness of the weight assignments, a sensitivity analysis was conducted by perturbing each weight by $\pm 10\%$ and recomputing the top-70 selection. Six perturbation scenarios were tested: (i) all weights increased by 10%, (ii) all weights decreased by 10%, (iii–iv) R1 (keyword match) varied $\pm 10\%$ individually, and (v–vi) R2 (citation impact) varied $\pm 10\%$ individually. The overlap between the original and perturbed top-70 lists ranged from 54.3% to 80.0% (mean 66.2%), with Jaccard similarity coefficients between 0.37 and 0.67 (mean 0.50). This indicates moderate robustness: while the exact composition of the top-70 list shifts under weight perturbation, the majority of papers remain stable, and the thematic coverage across all ten sub-themes is preserved. The observed variation is concentrated at the boundary of the selection threshold (papers with near-threshold scores), which is expected given the long-tail score distribution. The manual-adjustment rules (zero-citation override, theme-coverage cap) further stabilize the selection by ensuring that papers introducing unique thematic concepts are retained regardless of minor score fluctuations.

2.6. Inclusion and Exclusion Criteria

The full inclusion and exclusion criteria are documented below.

- A paper was included if it satisfied all of the following: (I1) topical relevance to at least one of the review's ten core themes; (I2) publication in a recognized academic venue indexed in Google Scholar with a DOI or named venue; (I3) abstract availability (mandatory); (I4) publication year preference 2015–2026, with foundational earlier works included if highly cited; (I5) English language; (I6) preference for reputable sources (Elsevier, IEEE, Springer, MDPI, ACM, Wiley, AAAI, SIGIR, etc.).
- A paper was excluded if any of the following applied: (E1) off-topic primary subject (medical, cybersecurity, education, HR, agriculture, manufacturing, physics, biology, etc.); (E2) no abstract; (E3) unverifiable (no DOI and no venue); (E4) duplicates; (E5) editorials, errata, or non-scholarly types; (E6) non-English without English metadata.

3. The 4S Lifecycle Framework

Based on the integrative synthesis of the 70 selected studies, this paper introduces the Signal–Structure–Surface–Score (4S) lifecycle framework (Figure 2). The framework captures the end-to-end process by which organizations must adapt their BI and SEO practices to the generative search era, mapping four sequential stages that bridge AI search mechanics with strategic measurement.

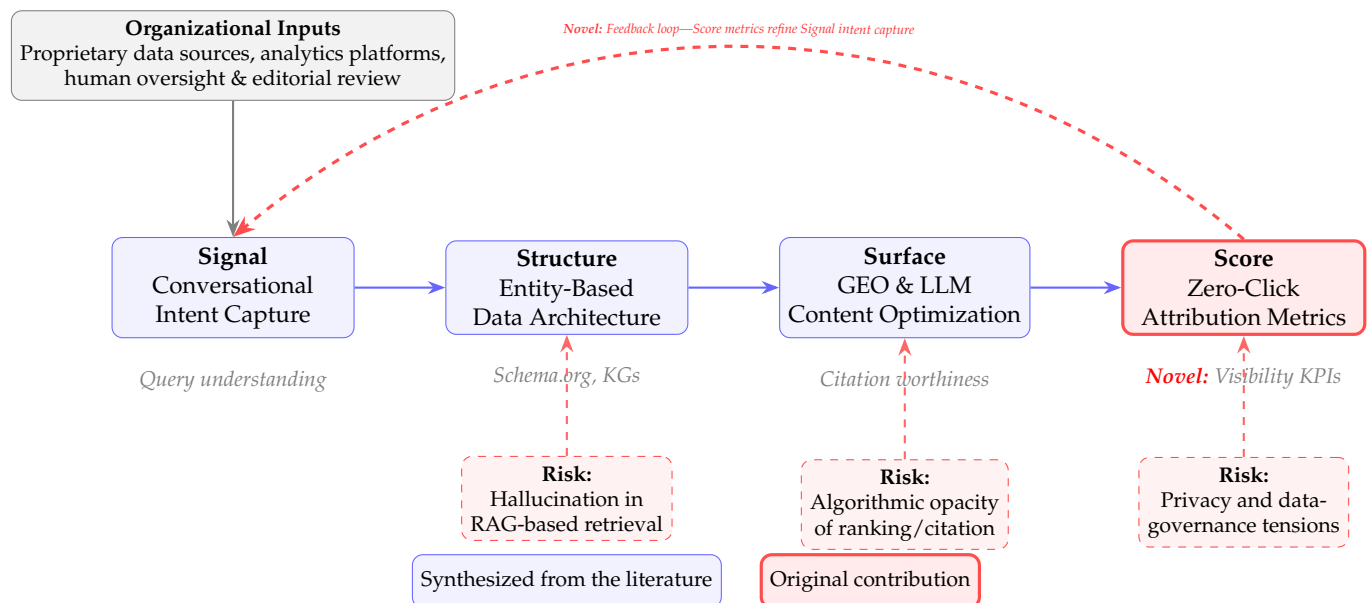


Figure 2. The Signal–Structure–Surface–Score (4S) lifecycle framework for AI-driven search visibility. Each stage represents a critical transition from traditional SEO to generative engine optimization, bridging AI search mechanics with BI measurement. Blue elements (Signal, Structure, Surface, the organizational-inputs node, and the risk nodes) are synthesized from prior literature; red elements (the Score stage and its feedback arrow) are original contributions of this review, as distinguished by the legend at the bottom. The dashed red feedback arrow indicates that Score-stage metrics (e.g., Share of Model, citation density) continuously refine Signal-stage intent capture, forming a closed organizational loop rather than a one-off linear process. The organizational-inputs node (**top**) represents the proprietary data sources, analytics platforms, and human oversight that feed the lifecycle, while the dashed red risk nodes (**bottom**) mark the three implementation risks—hallucination, algorithmic opacity, and privacy/data-governance tensions—identified in Sections 4.2 and 5.2 as recurring obstacles at the Structure, Surface, and Score stages, respectively.

3.1. Novelty and Contribution of the 4S Framework

To situate the 4S framework within the existing landscape and demonstrate its novelty, Table 2 compares it against five prior frameworks that span the three domains this review bridges: SEO process models, GEO/AEO positioning frameworks, and BI/analytics lifecycle models. The comparison reveals that no existing framework integrates all four capabilities that the generative search era demands: (a) upstream conversational intent capture, (b) entity-based data architecture for external LLM retrieval, (c) content optimization for citation-worthiness, and (d) zero-click attribution measurement with feedback loops to content strategy.

Building on the comparison in Table 2, three elements of the 4S framework are novel relative to all five prior models. First, the integration of BI measurement (Score) as a lifecycle stage: existing SEO and GEO frameworks either treat measurement as an afterthought (Prasad’s rank tracking [1]) or propose influence-oriented KPIs without specifying how organizations collect them at scale (Samet’s SoM [6]). The 4S framework embeds measurement as a co-equal lifecycle stage with explicit proxy-collection methods (manual auditing, third-party tools, synthetic benchmarking) and BI dashboard integration, addressing the black-box challenge that makes prior KPI proposals theoretical rather than operational. Second, the feedback loop between technical optimization and business metrics: all five prior frameworks are essentially linear—SEO funnels move from keyword research to rank tracking; GEO-First moves from Digital Authority to Discovery/Sentiment; Kimball’s BI lifecycle moves from planning to maintenance. The 4S framework’s dashed feedback arrow (Score → Signal) closes the loop, ensuring that visibility metrics continuously refine

intent capture and content strategy, formalizing the “discovery-to-branded-search loop” that Samet [6] describes but does not structurally embed in the lifecycle. Third, the explicit mapping to generative search mechanics: prior frameworks were designed for either the Retrieve-and-Rank paradigm (Prasad, Vollrath) or the GEO positioning layer (Samet, Enache) but do not trace the full pipeline from conversational query understanding (Signal) through entity-based data architecture (Structure) and LLM citation optimization (Surface) to zero-click attribution (Score). The 4S framework is the first to map this end-to-end pipeline, explicitly connecting RAG retrieval mechanics, knowledge graph architecture, and GEO content optimization to BI measurement within a single lifecycle.

Table 2. Comparison of the 4S lifecycle framework against prior SEO, GEO, and BI frameworks.

Framework	Stages	Key Limitation	How 4S Addresses It
Traditional SEO Process Model [1]	Keyword research → On-page optimization (8 targets) → Link building → Rank/CTR tracking	Built on the Retrieve-and-Rank paradigm; CTR-centric measurement; no concept of conversational intent or zero-click attribution	Signal replaces keyword research with conversational intent capture; Structure generalizes on-page to entity architecture; Score replaces CTR with SoM/citation density
GEO-First [6]	Digital Authority (foundation) → Discovery + Sentiment pillars; KPIs: Share of Model, citation density	Strategic positioning, not an operational lifecycle; no upstream intent or data-architecture stages; no BI collection pipeline for opaque APIs	Signal adds the missing intent-capture stage; Structure supplies the KG/Schema.org plumbing; Discovery assumes; Score operationalizes SoM within a BI governance pipeline with proxy-collection methods
SEO→AEO Transition [9]	Semantic Clarity → Schema.org markup → Extraction-Readiness	Narrowly scoped to content preparation; no intent-capture stage; no measurement or BI integration	Signal adds intent capture; Structure generalizes extraction-readiness to full KG architecture; Score adds the measurement layer connecting citation to business value
Digital Marketing Analytics/CDJ [8]	Strategy → Map analytics to Customer Decision Journey → Tool selection → Reconnect to ROMI	Measurement-governance only; predates generative search; assumes click/referral infrastructure; remedy is tool selection, not new KPI categories	Score directly operationalizes “reconnect measurement to strategy” for zero-click with SoM/citation density; feedback loop formalizes the closed-loop governance their framework calls for
Kimball BI Lifecycle	Planning → Requirements → Dimensional modeling → Physical design → ETL → BI apps → Deployment → Maintenance	Inward-facing warehouse engineering; assumes controlled internal data sources; no external-LLM-retrieval stage; no visibility-in-AI-answer KPIs	Signal + Structure extend modeling outward to conversational logs and entity content; Surface is the new external-retrieval-optimization stage; Score adapts ETL/dashboards to opaque generative-search data sources

The 4S framework emerged from the thematic analysis of the 70 selected papers through an iterative process. During the synthesis phase (Section 3), papers were grouped by their primary contribution: those addressing query understanding and conversational intent (e.g., [23–25]) clustered into the Signal stage; those addressing structured data, knowledge graphs, and Schema.org (e.g., [10,11,26]) clustered into the Structure stage; those addressing GEO, AEO, and citation-worthiness (e.g., [6,9,27]) clustered into the Surface stage; and those addressing zero-click metrics, attribution, and BI analytics (e.g., [6,8,28]) clustered into the Score stage. The feedback loop was identified when the synthesis revealed that Score-stage metrics (SoM, citation density) are not terminal outputs but inputs that inform subsequent content and markup decisions—a pattern documented across multiple reviewed studies but not formalized in any existing framework. The framework thus synthesizes prior work rather than replacing it: each stage draws on established literature (Signal on conversational search, Structure on KG/Schema.org research, Surface on GEO/AEO, Score on BI analytics), but the integration of all four into a closed lifecycle with feedback loops is the original contribution of this review.

3.2. Framework Overview

Figure 2 presents the complete 4S lifecycle framework, combining its operational detail with a visual distinction between original and synthesized contributions. Blue elements (the Signal, Structure, and Surface stages, the organizational-inputs node, and the risk nodes) denote stages and considerations synthesized from prior literature (conversational search, KG/Schema.org research, GEO/AEO, and known implementation risks), while red elements (the Score stage and its feedback arrow) denote the two original contributions of this review: the integration of BI measurement as a co-equal lifecycle stage, and the closed feedback loop from Score back to Signal that no prior SEO, GEO, or BI framework formalizes. A legend at the bottom of the figure distinguishes synthesized from original components.

Table 3 complements Figure 2 by detailing the concrete activities, data sources, KPIs, and validation criteria associated with each of the four 4S stages, translating the conceptual lifecycle into an operational reference for practitioners.

3.3. Signal: Conversational Intent Capture

The first stage addresses how AI-powered search engines interpret user queries. Unlike traditional keyword-based matching, generative search systems employ LLMs to understand conversational, multi-turn, and context-dependent queries [23,24]. The shift from keyword matching to intent understanding means that organizations must capture not just which keywords users type, but what they mean—a transition that requires NLP-driven intent classification and query rewriting capabilities [25]. Research on conversational search systems demonstrates that user query formulation strategies differ fundamentally between traditional and LLM-based search, with the latter producing longer, more natural-language queries [23]. This stage aligns with the BI “Signal” function: detecting and interpreting the conversational signals that generative engines use to retrieve and synthesize information.

To make this difference concrete, consider the following comparison drawn from the reviewed literature. In traditional keyword-based search, a user seeking information about optimizing content for AI search might type the terse query “AI search optimization”—a keyword string that the search engine matches against inverted indexes of page titles, body text, and anchor text, returning ranked blue-link results. The user then scans the result list, clicks on promising links, and self-synthesizes the answer [29]. In contrast, in a generative search environment powered by LLMs, the same user might phrase the query as a natural-language question: “How should I structure my website content so that AI search engines like Google SGE or Perplexity are more likely to cite it?” [23]. Sekuli

et al. [24] empirically demonstrate that users formulate longer, more semantically rich queries with LLM-based search than with traditional search engines—a behavioral shift confirmed by Mo et al. [23], who document the evolution from keyword-based to dialogue-based search. Ma et al. [25] show that query rewriting—reformulating such conversational queries before retrieval—significantly improves search relevance, because the LLM can disambiguate intent (e.g., distinguishing “optimization for AI search” from “using AI for search optimization”) and expand implicit references (e.g., resolving “SGE” to “Search Generative Experience”) that keyword matching would miss. Liang et al. [5] provide further empirical evidence through a user study comparing Bing Chat with traditional search, demonstrating that generative IR returns synthesized text with citations rather than ranked links, thereby reducing user search effort. For organizations, this means the “Signal” stage is fundamentally different from traditional keyword research: instead of tracking which keyword strings drive traffic, BI systems must capture and classify conversational intent—the underlying need behind a user’s natural-language query—and feed it into content strategy and RAG retrieval pipelines.

Table 3. Operational summary of the 4S lifecycle framework: activities, data sources, KPIs, and validation criteria for each stage.

Stage	Activities	Data Sources	KPIs	Validation Criteria
Signal	Query understanding, intent classification, conversational query rewriting	Search query logs, conversational AI transcripts, LLM search interaction logs	Query complexity/length trends, intent-classification accuracy, conversational query volume	Alignment between classified intent and actual user conversion path; inter-rater agreement on intent labels
Structure	Schema.org markup implementation, knowledge graph construction/enrichment, structured-data auditing	Website structured data (JSON-LD/microdata), internal knowledge bases, public KGs (e.g., Wikidata)	Schema.org markup coverage (%), KG entity completeness, structured-data validation pass rate	Schema validator (e.g., Google Rich Results Test) pass rate; KG entity resolution accuracy
Surface	Content restructuring for extraction-readiness, citation-worthiness optimization, AI content pipeline management	LLM answer logs, competitor citation audits, content performance analytics	Citation frequency in AI answers, extraction-readiness score, content production velocity	Manual audit of LLM citation accuracy; A/B testing of structured vs. unstructured content citation rates
Score	Zero-click KPI definition, BI dashboard integration, attribution model design	Third-party rank-tracking tools (e.g., SEMrush, Ahrefs), manual LLM-answer audits, synthetic query panels	Share of Model (SoM), citation density, brand mention frequency	Cross-validation of manual audits against automated citation-tracking tools; correlation with downstream conversion metrics

3.4. Structure: Entity-Based Data Architecture

The second stage involves architecting content as machine-readable, entity-based data. Traditional SEO relied on HTML structure and metadata tags; generative search requires structured data in the form of knowledge graphs, Schema.org markup, and linked data [10,11]. Research on knowledge graph augmentation with LLMs demonstrates that microdata embedded in web pages for SEO purposes can serve as a valuable source for enriching knowledge graphs [11], creating a bidirectional relationship between SEO practices and KG construction. The integration of business intelligence with knowledge

graph structures enables organizations to represent their products, services, and expertise in formats that LLMs can reliably retrieve and cite [9].

3.5. Surface: GEO and LLM Content Optimization

The third stage focuses on optimizing content for LLM retrieval and citation—the practice increasingly termed Generative Engine Optimization (GEO) or Answer Engine Optimization (AEO) [6,9]. Research shows that visibility in generative search results depends not on traditional ranking signals but on “citation worthiness”—the semantic clarity, authority, and structured format of content [30]. Samet [6] proposes a “GEO-First” framework that unifies search visibility, online reputation management, and digital authority, arguing that narrative inclusion within an AI’s retrieval set acts as a catalyst for subsequent high-intent branded searches. Enache [9] introduces a structured approach to transitioning from SEO to AEO, emphasizing Schema.org markup and “extraction-readiness” as prerequisites for LLM citation. AI-generated content pipelines further accelerate this process, with studies demonstrating 98.8% efficiency improvements in SEO content production through generative AI integration [27].

3.6. Score: Zero-Click Attribution Metrics

The final stage addresses the measurement challenge: how organizations define and track visibility in zero-click environments. Traditional SEO metrics—keyword rankings, organic traffic, click-through rates—lose meaning when search engines synthesize answers without directing users to external sites [28]. Research on digital marketing analytics highlights the risk of “analytics myopia”—over-reliance on legacy metrics that fail to capture the full customer decision journey [8]. Samet [6] proposes “Share of Model” (SoM) and citation density as novel KPIs for the AI search era, arguing that organizations must transition from click-oriented to influence-oriented metrics. This stage represents the BI “Score” function: defining the attribution frameworks that capture value in a landscape where clicks no longer serve as the primary visibility signal.

From a BI perspective, three interconnected challenges require explicit explanation. Attribution loss: Traditional CTR-based attribution models break down in zero-click environments because generative search engines synthesize answers in-session, often without directing users to external websites [28]. When Google SGE, ChatGPT, or Perplexity cite a source within an AI-generated answer, the referring-site mechanism that powered web analytics for two decades is absent—no HTTP referrer, no UTM parameter, no landing-page visit. BI alternatives include impression-based attribution (tracking how often a brand or content appears in AI-generated answers), citation-based attribution (counting the frequency and context of LLM citations), and influence-based attribution (measuring downstream branded-search or direct-traffic lifts correlated with AI answer presence) [6]. Zero-click metrics: Organizations require concrete KPIs tailored to generative search. Samet [6] proposes “Share of Model” (SoM)—the proportion of AI-generated answers that reference a given brand relative to competitors—as a primary visibility metric. Citation density (the number of citations per answer or per query), brand mention frequency (how often a brand appears across a panel of representative queries), and answer sentiment polarity (whether the LLM frames the brand positively or negatively) complement SoM as a composite KPI suite. These metrics are analogous to share-of-voice in traditional media measurement but adapted for generative retrieval. Data governance: Capturing generative search signals requires organizations to restructure their data pipelines. Traditional SEO dashboards ingest server-log data, Google Search Console exports, and third-party rank-tracking feeds; generative search demands new data sources—LLM answer logs, synthetic query panels, and manual audit records—that existing BI pipelines were not designed

to handle. Organizations must establish governance protocols for auditing AI-generated answers at scale, define sampling strategies for representative query sets, and integrate generative-search metrics alongside legacy SEO KPIs in unified BI dashboards [8]. This data-governance restructuring is both a technical challenge (pipeline architecture, data schema design) and an organizational challenge (defining ownership, audit cadence, and reporting standards for metrics that lack industry-standard definitions).

A critical practical limitation of the Score stage must be acknowledged: Google SGE, OpenAI, Perplexity, and other generative search providers do not currently expose public APIs for tracking when, how, or whether a given source is cited within AI-generated answers. Unlike traditional search, where Google Search Console and third-party rank-tracking APIs (e.g., SEMrush, Ahrefs, Moz) provide structured programmatic access to impression, click, and ranking data, generative search engines operate as closed systems whose citation and retrieval mechanisms are opaque to external observation. This black-box challenge means that the zero-click KPIs proposed above—Share of Model, citation density, brand mention frequency—cannot, at present, be collected automatically at scale through a standardized API; they must instead be gathered through proxy methods. Three complementary proxy approaches are available. Manual auditing: Organizations can construct representative query panels—curated sets of domain-relevant queries—and periodically submit them to Google SGE, ChatGPT, Perplexity, and similar engines, manually recording which sources are cited, in what position, and with what framing, while labor-intensive, this method yields the most accurate citation-level data and can be systematized through scripted query submission combined with human annotation. Third-party tools: Emerging SEO platforms (e.g., Profound, Otterly.AI, AthenaHQ) are beginning to offer generative-search tracking features that automate the submission of query panels and aggregate citation data across multiple LLM-based engines; established tools such as SEMrush and Ahrefs are also extending their rank-tracking capabilities to include AI answer panel visibility. These tools provide a scalable proxy for Share of Model and citation density, though their coverage is limited to the engines and query types they support. Synthetic benchmarking: Organizations can build internal RAG-based search prototypes using their own content corpora and open-source LLMs (e.g., Llama, Mistral) to simulate generative retrieval, measuring how frequently their content is selected and cited under controlled conditions, while not equivalent to production generative search, synthetic benchmarking provides a reproducible, low-cost method for testing the impact of Schema.org markup, content structure, and entity coverage on citation likelihood. The black-box challenge is both a limitation and a research opportunity. As a limitation, it means that current Score-stage metrics are necessarily approximate and labor-intensive, and that cross-engine benchmarking remains inconsistent because each generative search system uses a different retrieval and citation mechanism. As a research opportunity, the absence of standardized citation-tracking APIs creates a clear opening for the development of open auditing frameworks—analogue to the ranking-transparency tools that emerged for traditional search—that could provide the infrastructure for rigorous, reproducible generative-search visibility measurement. This need is explicitly reflected in the Future Research Agenda (Section 5.4), which calls for the development of zero-click attribution frameworks and algorithmic transparency methods.

4. Literature Review

The 70 selected papers are organized into eight thematic categories that collectively map the intersection of SEO, generative AI, and business intelligence. Rather than treating these categories in isolation, this review traces the connections between them—how technical advances in retrieval architecture enable new marketing strategies, how knowledge

graph research informs entity-based SEO, and how bias and fairness concerns shape the future of search visibility measurement.

Table 4 provides an explicit mapping of the eight thematic categories onto the four stages of the 4S lifecycle framework, showing which categories contribute evidence to which stages. A checkmark (✓) indicates primary evidence; a dash (—) indicates no direct contribution.

Table 4. Theme-to-4S evidence matrix mapping the eight thematic categories onto the four stages of the 4S lifecycle framework. ✓ = primary evidence; — = no direct contribution.

Thematic Category	Signal	Structure	Surface	Score
Section 4.1 SEO Evolution and GEO	—	✓	✓	✓
Section 4.2 RAG and Information Retrieval	✓	✓	—	✓
Section 4.3 Knowledge Graphs and Semantic Search	—	✓	✓	—
Section 4.4 AI-Driven Digital Marketing	✓	—	—	✓
Section 4.5 Conversational AI and Chatbots	✓	✓	—	—
Section 4.6 Bias, Fairness, and Ethics in IR	—	—	—	✓
Section 4.7 Sentiment Analysis and Customer Behavior	✓	—	—	✓
Section 4.8 Business Intelligence and Predictive Analytics	—	—	—	✓

No single thematic category addresses all four stages, but together they provide complete coverage of the 4S lifecycle. The Signal stage is supported by categories that capture user intent and conversational behavior (Sections 4.2, 4.4, 4.5 and 4.7); the Structure stage by categories that define data architecture and retrieval mechanisms (Sections 4.1–4.3 and 4.5); the Surface stage by categories that address content optimization for LLM citation (Sections 4.1 and 4.3); and the Score stage by categories that define measurement, governance, and analytics (Sections 4.1, 4.2, 4.4 and 4.6–4.8).

4.1. SEO Evolution and Generative Engine Optimization

The most striking finding across this body of literature is the speed and depth of the SEO paradigm shift. What was once a discipline built on keyword matching, back-link profiles, and on-page technical optimization [1,31] has fractured into a multi-layered practice where visibility depends not on ranking position but on whether a generative AI system chooses to cite a source. Liang et al. [5] provide empirical evidence for this transition through a user study comparing Bing Chat (a generative IR system) with traditional search, demonstrating that generative IR returns synthesized text with citations rather than ranked links, thereby reducing user search effort and fundamentally altering the click-based economy of organic search.

This observation is not merely theoretical. Beriozkin [32] provides concrete evidence from a niche luxury transportation firm, where the deployment of Custom GPTs enriched with real-time SEO analytics produced a 23% increase in organic traffic and a 37% uplift in lead conversion rates. What makes this case instructive is not just the positive outcomes but the mechanism: the firm transcended the limitations of generic LLMs by training AI agents on proprietary corporate data and external keyword metrics, effectively creating a localized knowledge base that generative engines could retrieve and cite. This finding suggests that the future of SEO may lie less in optimizing for search engine algorithms and more in optimizing for LLM retrieval—a shift that Samet [6] formalizes as a “GEO-First” framework.

Samet’s framework is particularly significant because it reframes the strategic question. Rather than asking “how do we rank?”, organizations should ask “how do we become cited?” The framework identifies two operational pillars—Discovery and Sentiment—grounded in a foundational layer of Digital Authority, and introduces “Share of Model” (SoM) as a novel KPI measuring how frequently an AI system references a brand. This repre-

sents a fundamental departure from the click-oriented metrics that have dominated digital marketing for two decades [4], aligning instead with what Samet terms “influence-oriented KPIs” suited to a zero-click, agentic landscape. Rejón-Guardia et al. [33], conversely, provide a complementary encyclopedia-level treatment of GEO, defining it as the strategies developed by marketing professionals to optimize content so that it is selected and presented by AI systems, with greater emphasis on semantic relevance, content quality, and adherence to E–E–A–T principles (experience, expertise, authoritativeness, and trustworthiness), as well as using structured data, optimization for conversational queries, and continuous content updating. Their work confirms that GEO constitutes a key strategic evolution from conventional SEO, with significant implications for visibility, web traffic, and competitive research.

The transition from SEO to AEO requires not just strategic reframing but technical adaptation. Enache [9] provides a methodological bridge, proposing that content must be repositioned as a “structured, machine-interpretable data object” with Semantic Clarity, Schema.org markup, and “Extraction-Readiness” as prerequisites for LLM citation. This architectural perspective is complemented by Salem et al. [10], who demonstrate that automatically injecting structured data into news articles—validated through Google’s Rich Results Test API—can improve discoverability for aggregator platforms. Together, these studies suggest that the technical SEO toolkit is expanding from HTML optimization to semantic web integration, where the ability to produce machine-readable, entity-based content becomes a core competency.

The content production process itself is being transformed. Murdiyanto et al. [27] demonstrate an end-to-end pipeline integrating web scraping, generative AI (OpenAI GPT), and autonomous WordPress publication, achieving a 98.8% reduction in content production time (from 195 min to 2.25 min per article) while meeting on-page SEO indicators. Vajrobol et al. [34] frame this transformation more broadly, arguing that generative AI is not merely accelerating SEO but fundamentally reshaping it—automating keyword research, enabling chatbot-enhanced user experiences, and providing data-driven insights that were previously inaccessible. However, both studies caution that AI-generated content raises ethical questions about quality, authenticity, and the potential for search engines to penalize machine-generated text.

A critical insight emerges when these findings are read together: the SEO profession is bifurcating. On one branch, technical practitioners must master structured data, knowledge graphs, and semantic web technologies to ensure content is “extraction-ready” for LLM retrieval [9,10]. On the other, content strategists must navigate an AI-accelerated production environment where the line between human and machine authorship is increasingly blurred [27,34]. The traditional on-page optimization factors identified by Prasad and Chandrika [1]—keyword optimization, site structure, internal linking, usability, mobile versioning, rich snippets, and social media integration—remain necessary but are no longer sufficient. They must be augmented with entity-based SEO, Schema.org markup, and the kind of digital authority signals that Samet [6] identifies as foundational to generative search visibility.

Synthesizing these findings, this body of literature maps primarily to the Surface and Score stages of the 4S framework: GEO and AEO practices define how content must be structured for LLM retrieval (Surface), while the proposed influence-oriented KPIs—Share of Model, citation density—introduce the measurement paradigms that the Score stage requires. The bifurcation of the SEO profession mirrors the bifurcation of the 4S lifecycle: technical practitioners operate in the Structure and Surface stages (structured data, extraction-readiness), while strategists operate in the Signal and Score stages (intent capture, visibility measurement).

4.2. Retrieval-Augmented Generation and Information Retrieval

The technical foundation of generative search lies in retrieval-augmented generation (RAG), a paradigm that has evolved rapidly from a niche technique to the dominant architecture for LLM-powered search systems. The literature reveals a clear trajectory: from foundational surveys that established the RAG taxonomy [15], through comprehensive frameworks that mapped its architectural variations [12,13], to specialized techniques that address specific retrieval challenges. The infrastructure supporting these advances—big data, cloud computing, and edge computing—plays a critical role in enabling the scalable NLP capabilities that generative search demands [35].

Gao et al. [15] provide the foundational survey that subsequent studies build upon, establishing the basic RAG framework of retrieving external knowledge to augment LLM generation. Huang and Huang [12] extend this foundation by organizing the RAG paradigm into four categories—pre-retrieval, retrieval, post-retrieval, and generation—offering a lifecycle perspective that reveals where optimization interventions can be applied. Fan et al. [13] further systematize the field across three dimensions: architectures, training strategies, and applications, demonstrating that RA-LLMs have moved beyond experimental prototypes into production deployment across diverse domains.

What emerges from these surveys is a picture of a maturing field grappling with fundamental reliability challenges. Chen et al. [36] establish the Retrieval-Augmented Generation Benchmark (RGB), evaluating six representative LLMs across four abilities: noise robustness, negative rejection, information integration, and counterfactual robustness. Their findings are sobering: while LLMs exhibit some noise robustness, they struggle significantly with negative rejection (failing to ignore irrelevant retrieved passages), information integration (combining multiple sources coherently), and handling false information. Zhang et al. [37] review hallucination mitigation techniques, confirming that hallucination remains a critical challenge for the reliability of generative search systems—one that RAG mitigates but does not eliminate.

The field's response to these challenges has been a proliferation of specialized retrieval strategies. Sawarkar et al. [38] propose “Blended RAG,” combining dense vector indexes with sparse encoder indexes and hybrid query strategies, setting new benchmarks on IR datasets and surpassing fine-tuning performance on generative Q&A tasks. Omolayo [39] contributes a comparative evaluation of vector embedding frameworks, providing practical guidance for selecting retrieval infrastructure. Su et al. [40] introduce DRAGIN, which dynamically triggers retrieval based on real-time LLM information needs rather than fixed query intervals, while Jeong et al. [41] propose Adaptive-RAG, which adjusts retrieval strategy based on question complexity—simple queries receive lightweight retrieval, complex queries invoke multi-step retrieval. Ma et al. [25] address the critical preprocessing step of query rewriting, showing that reformulating user queries before retrieval significantly improves search relevance in conversational contexts.

A particularly important finding for the SEO community comes from Li et al. [42], who compare RAG with long-context LLMs and propose a hybrid approach. Their study suggests that the choice between retrieving external knowledge and providing longer context windows is not binary—optimal systems will combine both, with implications for how content should be structured and chunked for retrieval. Sun et al. [43] directly investigate whether ChatGPT is effective as a search engine, examining LLMs as re-ranking agents. Their finding that LLM performance varies significantly across query types is a critical insight for SEO practitioners: generative search does not uniformly advantage all content types, and understanding which query classes LLMs handle well or poorly can inform content strategy.

The broader IR perspective is provided by Zhu et al. [44], who survey the confluence of LLMs and IR systems across query rewriters, retrievers, rerankers, and readers, and by Hersh [29], who argues that search still matters in the era of generative AI—information retrieval remains a fundamental capability even as generative systems evolve. Mo et al. [23] map the evolution from keyword-based to dialogue-based search in their survey of conversational search, while Sekuli et al. [24] analyze utterances in LLM-based user simulation, revealing that users formulate longer, more natural-language queries with LLM-based search than with traditional search engines—a behavioral shift with profound implications for keyword research and content strategy.

Table 5 summarizes the key RAG variants discussed in this section, mapping each against its query complexity mechanism, retrieval strategy, and implication for content layout and data structures.

Table 5. Comparative summary of RAG variants reviewed in Section 4.2.

RAG Variant	Query Complexity Mechanism	Retrieval Strategy	Implication for Content Layout/Data Structures	Key Citation
Blended RAG	Static (fixed retrieval step)	Blended (dense + sparse)	Content must support both semantic vector matching and lexical BM25-style retrieval; hybrid indexing requires clean text structure and keyword-rich passages	[38]
Adaptive-RAG	Adaptive (complexity-based)	Dynamic (lightweight or multi-step based on query type)	Content should be modular: simple factual content for lightweight retrieval, multi-source synthesis content for complex queries; chunk size matters	[41]
DRAGIN	Dynamic (real-time LLM-triggered)	Dense (triggered on-demand during generation)	Content must be chunked into semantically coherent units that can be retrieved mid-generation; entity-level granularity preferred over document-level	[40]
Hybrid RAG + Long-Context	Hybrid (retrieval + extended context window)	Blended (external retrieval + internal long-context)	Content should be structured for both chunked retrieval and full-context processing; redundancy across chunks helps when content is consumed via long-context windows	[42]

Taken together, the RAG and IR literature underpins the Signal and Structure stages of the 4S framework. Query rewriting and conversational search research informs how organizations must capture conversational intent (Signal), while the proliferation of retrieval strategies—blended, adaptive, dynamic—defines the technical mechanisms through which structured content is retrieved and cited by LLMs (Structure). The reliability challenges documented by Chen et al. [36] and Zhang et al. [37] also feed directly into the Score stage: organizations must account for hallucination risk when measuring the accuracy of their LLM-generated citations.

4.3. Knowledge Graphs and Semantic Search

If RAG provides the retrieval engine for generative search, knowledge graphs provide the fuel. The literature reveals a symbiotic relationship between SEO-optimized structured data and knowledge graph construction—one that creates new strategic opportunities for organizations willing to invest in entity-based content architecture.

Gonzalez-Garcia et al. [11] demonstrate this symbiosis empirically. By extracting Schema.org microdata from web pages via CommonCrawl and matching it against Wikidata, they show that microdata produced for SEO purposes can serve as a valuable resource for knowledge graph enrichment. The implication is striking: SEO practitioners have been producing structured data for years without realizing its potential value for KG construction. Conversely, KGs enhance search engine understanding of web content, creating a virtuous cycle where SEO investment in structured data improves KG quality, which in turn improves search visibility. Their finding that metadata annotations concentrate in only a few Schema.org attributes—and that authoring tools influence a significant fraction of microdata production—suggests both an opportunity and a quality challenge for the SEO community.

Ibrahim et al. [26] provide the comprehensive survey that maps how LLMs can augment KGs across construction, completion, and question answering, while Wang et al. [45] demonstrate the reverse: using KGs to augment LLM-based recommendation through Knowledge Graph Retrieval-Augmented Generation. This bidirectional enhancement—LLMs improving KGs and KGs improving LLMs—represents a convergence that is reshaping both fields. The practical question for organizations is whether to invest in building proprietary KGs or in optimizing content for inclusion in existing ones.

Sun et al. [46] provide a provocative answer with their “Head-to-Tail” study, which assesses LLM knowledgeability across different frequency bands. Their finding that LLMs excel at common (“head”) knowledge but struggle with rare (“tail”) knowledge suggests that KGs remain essential for specialized information—and that organizations with deep domain expertise can establish authority by structuring that expertise in KG format. This finding has direct SEO implications: in a generative search environment, being the authoritative source for niche entities that LLMs cannot reliably generate from training data may be more valuable than competing for common knowledge.

The technical foundations for this entity-based approach are well-established. Xie et al. [47] present foundational work on representation learning of KGs with entity descriptions, a technique that underpins modern semantic search. Choi et al. [48] demonstrate how pre-trained language models can enhance KG completion through masked entity prediction, while Wang et al. [22] address explainable reasoning over KGs for recommendation, contributing to the transparency that both search engines and regulators increasingly demand.

The semantic search tradition that predates the generative AI boom remains theoretically relevant. Janowicz et al. [49] examine the semantics of similarity in geographic information retrieval, providing theoretical foundations that inform modern GEO practices. Hollink et al. [50] conduct semantic search log analysis on professional image search, offering early insights into how users interact with semantic search systems—insights that are now being revisited as generative search transforms user behavior [24].

To make the connection between knowledge graphs and the present research process explicit, Figure 3 visualizes the end-to-end pipeline through which SEO-produced structured data flows into knowledge graph construction and, ultimately, into LLM retrieval and citation. The pipeline comprises three stages: (i) Schema.org markup embedded in web pages (JSON-LD, microdata) defines entities and their relationships in machine-readable form; (ii) these annotations are harvested—via large-scale crawls such as Common-Crawl [11] or via proprietary ingestion pipelines—and matched against existing knowledge graphs (e.g., Wikidata), enriching the graph with new entities and relations; (iii) generative search engines (Google SGE, Perplexity, ChatGPT) query the enriched knowledge graph during retrieval-augmented generation, selecting and citing authoritative sources whose entities match the user’s conversational intent. The dashed feedback arrow indicates that LLM citation outcomes (which sources are cited, for which queries) can be monitored by organizations to refine their Schema.org markup strategy, closing the loop between the Structure and Score stages of the 4S framework.

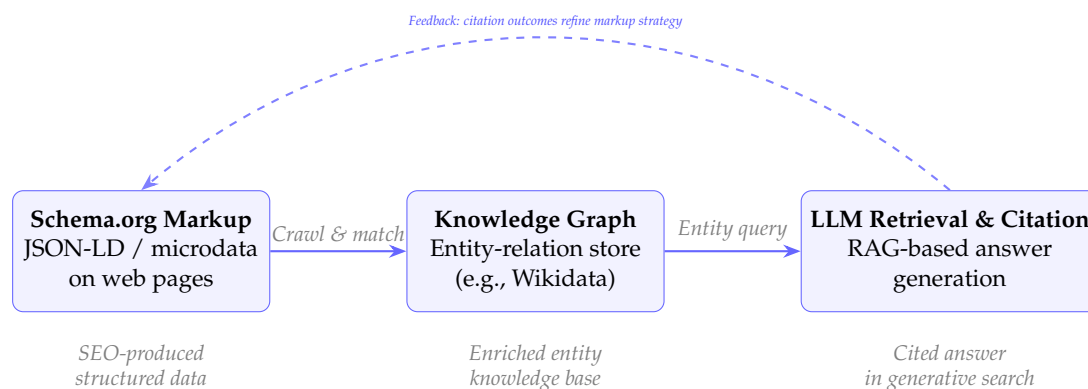


Figure 3. Knowledge graph pipeline for generative search visibility. SEO-produced Schema.org markup (**left**) is crawled and matched against existing knowledge graphs (**center**), which generative search engines query during retrieval-augmented generation to produce cited answers (**right**). The dashed feedback arrow indicates that citation outcomes inform markup refinement, closing the Structure–Score loop of the 4S framework.

Table 6 further structures the knowledge graph landscape by mapping KG types to their construction methods, SEO/GEO applications, and the key papers from the reviewed corpus that document each type.

Viewed through the 4S lens, the knowledge graph literature maps centrally to the Structure stage: KGs provide the entity-based data architecture that LLMs retrieve and cite. The bidirectional enhancement documented by Ibrahim et al. [26] and Wang et al. [45]—LLMs improving KGs and KGs improving LLMs—directly supports the Surface stage as well, since Schema.org markup and structured data produced for SEO purposes enhance the knowledge graphs that generative engines rely on for citation decisions.

Table 6. Knowledge graph types, construction methods, SEO/GEO applications, and key papers from the reviewed corpus.

KG Type	Construction Method	SEO/GEO Application	Key Papers
Public/encyclopedic KGs (e.g., Wikidata, DBpedia)	Crowdsourced entity editing; automated extraction from Wikipedia and structured web sources	Serve as the entity reference layer that generative engines query; organizations optimize content for entity inclusion and disambiguation	[11,49]
SEO-derived KGs (from Schema.org microdata)	Large-scale crawling of web pages (e.g., CommonCrawl); extraction of JSON-LD/microdata; matching against existing KGs	SEO practitioners produce structured data for ranking purposes that simultaneously enriches KGs; creates a virtuous cycle between SEO investment and KG quality	[10,11]
LLM-augmented KGs	LLM-based entity extraction, relation prediction, and KG completion (e.g., masked entity prediction)	Automate KG construction and completion, reducing the manual cost of entity curation; enable organizations to build proprietary KGs from unstructured internal content	[26,48]
KG–RAG hybrid KGs	Knowledge Graph Retrieval-Augmented Generation; KGs used as structured retrieval source alongside dense vector indexes	Improve LLM answer accuracy and citation provenance; organizations with proprietary KGs gain citation advantage for niche (“tail”) entities that LLMs cannot generate from training data alone	[45,46]
Explainable KGs	Reasoning over KGs with explainable paths; attention-based entity-path visualization	Provide transparency for ranking and citation decisions; support XAI requirements that regulators and practitioners increasingly demand	[22,47]

4.4. AI-Driven Digital Marketing and Personalization

The marketing literature reveals a field in rapid transformation, where AI is simultaneously a tool for optimization and a force reshaping the competitive landscape. A cross-cutting insight from this body of work is that AI adoption in digital marketing is no longer a differentiator—it is becoming a baseline requirement, and the competitive advantage lies in how deeply organizations integrate AI into their strategic decision-making.

Islam et al. [17] provide one of the most comprehensive assessments, using PRISMA methodology to synthesize 150 relevant studies. Their finding that AI enhances digital marketing through predictive analytics, NLP, and chatbots—enabling customer segmentation, content personalization, and campaign optimization—is consistent with the broader literature. What distinguishes their contribution is the identification of persistent challenges: data privacy, algorithmic bias, and high AI adoption costs remain significant barriers, particularly for smaller organizations. These challenges are echoed by Hasan [51], whose PRISMA-guided review of 112 articles confirms that AI-powered SEO tools improve organic reach, technical site health, and semantic keyword alignment, but notes that sustained AI adoption requires compounding strategic advantages including innovation capacity, brand loyalty, and agility.

The personalization dimension is explored by several studies that collectively reveal a shift from segment-based to individual-level marketing. Das [19] emphasizes generative

AI's capacity to analyze consumer data and produce tailored text, images, and video, highlighting the growing significance of voice and image search optimization. Raji et al. [52] review AI-powered personalization in e-commerce, showing how advanced algorithms deliver tailored content and recommendations while predicting consumer preferences and streamlining the purchasing journey. Kedi et al. [53] focus on SMEs, finding that machine learning, NLP, and predictive analytics enable personalized content delivery that was previously accessible only to large enterprises with dedicated data science teams.

The e-commerce context receives particular attention. Kathiriya [54] demonstrates LLM-based description and keyword generation from multimodal data for e-commerce listings, showing how generative AI can automate product content creation at scale. Purnomo [55] identifies SEO, content marketing, social media, paid advertising, and user experience optimization as core strategies for sales conversion, while Shaban [56] provides a comprehensive review of digital marketing strategies for e-commerce success. The regional dimension is added by Kamkankaew et al. [57], who examine AI's transformative role in Thai B2C digital marketing, providing evidence that AI-driven marketing transformation is not confined to Western markets.

A particularly interesting finding comes from Abdelkader [58], who investigates ChatGPT's influence on customer experience in digital marketing and finds that familiarity with technology, business type, age, and education level moderate the relationship between ChatGPT-based customer experience and overall satisfaction. This suggests that the effectiveness of AI-driven marketing is not uniform across customer segments—a critical insight for personalization strategies that must account for varying levels of AI acceptance. Arslan et al. [59] extend this line of inquiry with a comprehensive review of LLM applications in business and management, combining bibliometric analysis with thematic synthesis to show a post-2023 shift from general-purpose conversational use toward domain-specific and architecture-aware implementations that bridge the gap between AI capabilities and marketing practice.

The SME perspective is particularly important given the resource constraints these organizations face. Sharabati et al. [60] find that digital marketing strategies—including SEO, social media marketing, and customer engagement through digital channels—are essential drivers of digital transformation and economic results for SMEs. Ali and Gutiérrez [61] emphasize that companies adopting AI and personalized marketing gain clear competitive advantages, while Singh [62] provides a systematic review of digital marketing strategies and consumer engagement, examining emerging trends and measurement frameworks. Hutsaliuk and Mirzoiev [63] highlight the integration of AI, big data analytics, automation, and social media in modern marketing strategies, while Joel [64] examines the broader impact of digital transformation on business development strategies. Sihotang [65] proposes a digital marketing strategy using customer journey analysis, and Fadnavis [66] explores personalized marketing through machine learning—together painting a picture of a marketing discipline that is becoming increasingly data-driven, AI-augmented, and customer-centric.

From a lifecycle perspective, the digital marketing literature connects to the Signal and Score stages. Personalization and customer behavior research informs how organizations capture and classify consumer intent (Signal), while the persistent challenges of data privacy, algorithmic bias, and AI adoption costs identified by Islam et al. [17] and Hasan [51] shape the governance frameworks that the Score stage must address. The shift from segment-based to individual-level marketing mirrors the shift from keyword-based to intent-based search.

4.5. Conversational AI and Chatbots in Search

The evolution from intent-based chatbots to LLM-powered conversational agents represents a microcosm of the broader search transformation. Singh and Namin [67] provide a comprehensive survey on chatbots and large language models, offering a taxonomy of testing and evaluation techniques that reveals how LLMs have fundamentally expanded conversational agent capabilities. The significance of this evolution for search lies in the fact that conversational interfaces are becoming the primary mode of interaction with information retrieval systems—a trend that Mo et al. [23] document in their survey of conversational search.

The practical implementation challenges of conversational search are addressed by Pokhrel et al. [68], who demonstrate a RAG-based website chatbot combining web scraping, vectorization, and semantic search. Their system retrieves relevant document segments based on user queries and generates contextually appropriate responses, illustrating how the RAG architecture discussed in Section 4.2 can be operationalized in a conversational interface. Mazumder [69] extends this work to library search systems, showing that RAG-based search can enhance information discovery in specialized contexts beyond commercial search engines—a finding with implications for enterprise search and internal knowledge management.

At its core, the conversational AI literature directly informs the Signal stage: chatbots and LLM-powered conversational agents are the primary interface through which users express conversational intent in generative search. The RAG-based chatbot implementations demonstrated by Pokhrel et al. [68] also connect to the Structure stage, as they require structured, retrievable content to function effectively.

4.6. Bias, Fairness, Transparency, and Ethics in Information Retrieval

As generative AI becomes embedded in search infrastructure, the question of who gets seen and why takes on new urgency. The literature reveals that bias in IR systems is not a peripheral concern but a structural feature that generative AI may amplify rather than mitigate.

Dai et al. [70] provide the most comprehensive treatment, unifying bias and unfairness in IR systems as distribution mismatch problems. Their framework examines bias across three stages of LLM integration—data collection, model development, and result evaluation—and proposes mitigation strategies through distribution alignment. What makes their contribution particularly significant is the recognition that LLM integration introduces new bias pathways that did not exist in traditional IR systems, requiring fundamentally new fairness frameworks.

Bernard and Balog [71] conduct a systematic review of fairness, accountability, transparency, and ethics in IR, revealing a field struggling with definitional pluralism. Their finding that no standard definitions exist for these concepts—due to their multi-dimensional nature—highlights a governance gap: organizations cannot ensure fairness in generative search if they cannot define what fairness means in this context. They develop taxonomies of requirements and propose practical definitions, but the implementation challenge remains.

The practical consequences of search bias are demonstrated by Feng and Shah [72], who investigate gender bias in image search using adversarial attack queries. Their experiments on Google, Baidu, Naver, and Yandex reveal that adversarial attacks can trigger high levels of gender bias even in systems that have been corrected for such bias—a finding that underscores the fragility of bias mitigation in opaque, proprietary search systems. They design three re-ranking algorithms (epsilon-greedy, relevance-aware swapping, and fairness-greedy) to mitigate bias, providing actionable techniques that could be applied to generative search systems.

Castillo [73] provides foundational principles for fairness and transparency in ranking that remain relevant as generative search systems increasingly determine information visibility. The ethical dimension extends beyond bias to privacy. Saura et al. [74] investigate the ethics of AI-based digital marketing, identifying a “data privacy paradox” where the most profitable AI-driven marketing actions—cross-device tracking, real-time monitoring, behavioral analytics—are precisely those least connected to privacy and ethical standards. Their finding of a strong proximity between real-time tracking, IoT, and surveillance variables underscores the critical need to ethically understand how user behavior is monitored in AI-driven marketing environments.

Critically, the bias and ethics literature cuts across all four stages but is most consequential for the Score stage: organizations cannot measure visibility fairly if the generative search systems they audit are themselves biased. The frameworks proposed by Dai et al. [70] and Bernard and Balog [71] provide the governance foundation for the Score stage, while the privacy concerns identified by Saura et al. [74] shape the data-governance challenges of capturing generative search signals.

4.7. Sentiment Analysis and Customer Behavior

Sentiment analysis serves as a critical bridge between understanding customer behavior and informing search and marketing strategy. The literature reveals that generative AI is transforming this field as profoundly as it is transforming search itself.

Krugmann and Hartmann [75] provide the benchmark study, comparing GPT-3.5, GPT-4, and Llama 2 against established transfer learning models for sentiment analysis. Their finding that LLMs can surpass traditional methods in classification accuracy—despite their zero-shot nature—represents a significant shift for marketing research. Equally important is their finding that linguistic features such as lengthy, content-laden words improve classification performance while single-sentence reviews and unstructured social media text reduce it, providing practical guidance for when LLM-based sentiment analysis is appropriate. Chan and Im [76] contribute a complementary technical perspective, proposing a stacked CARU network with a Naïve-Bayes classifier for paragraph-level sentiment analysis, demonstrating that combining CNN-based feature extraction with probabilistic classification can effectively address long-term dependency problems in long sequences—a challenge that LLM-based approaches also face when processing extended review text.

The multilingual dimension is addressed by Miah et al. [77], who achieve over 86% accuracy on cross-lingual sentiment analysis using an ensemble of transformers and LLMs with translation to English. This finding has significant implications for global digital marketing strategies: organizations can leverage LLM-based sentiment analysis across language barriers without requiring language-specific training data. Rane et al. [78] review the broader application of AI, ML, and deep learning for sentiment analysis in CRM, highlighting how NLP-driven sentiment analysis enables real-time understanding of customer emotions across social media, reviews, and support tickets. Paul et al. [79] extend this to digital marketing specifically, finding that transformer-based models (BERT, RoBERTa) excel in multilingual sentiment interpretation and that integrating these models into customer feedback loops enables real-time marketing decision-making and sentiment-driven content optimization.

From a 4S perspective, the sentiment analysis literature enriches the Signal stage: understanding customer sentiment enables organizations to capture not just what users search for, but how they feel about the results—a richer intent signal than keyword data alone provides. The demonstrated superiority of LLMs over traditional methods for sentiment classification [75] also has implications for the Score stage, as sentiment polarity in AI-generated answers becomes a relevant KPI.

4.8. Business Intelligence and Predictive Analytics

The BI and predictive analytics literature addresses the measurement dimension of the 4S framework—how organizations capture, analyze, and act on data in an AI-driven search environment. A key insight from this body of work is that the tools and frameworks developed for traditional digital marketing analytics are necessary but insufficient for the generative search era.

Vollrath and Villegas [8] provide the conceptual foundation with their work on avoiding “digital marketing analytics myopia.” Their framework places analytics tools in the context of a firm’s overall marketing plan, reconnecting measurement with the customer decision journey. This principle is directly applicable to the zero-click attribution challenge: organizations that focus solely on declining click metrics risk missing the broader influence that generative search visibility exerts on brand awareness, consideration, and downstream conversion. The framework’s emphasis on theoretically grounded, strategically aligned measurement provides a template for developing the “Score” stage of the 4S lifecycle.

Ajiga et al. [80] review AI-driven predictive analytics in retail, demonstrating how machine learning algorithms, NLP, and computer vision revolutionize the way retailers harness data for strategic decision-making. Their emphasis on data quality and ethical considerations is a recurring theme: the value of predictive analytics is contingent on the quality of input data and the transparency of the models used. Cvitanović [81] provides early evidence of how BI dashboards support marketing decision-making through digital marketing benchmarks, while Chen et al. [82] demonstrate how RAG-based systems can serve as BI tools for organizational knowledge management, bridging the gap between internal knowledge and AI-powered retrieval.

The application of RAG to specialized BI domains is explored by Iaroshev et al. [83], who evaluate RAG models for financial report question answering, and by Toro et al. [84], who present DRAGON-AI for dynamic ontology generation. Together, these studies suggest that the RAG architecture—originally developed for general-purpose question answering—is being adapted for domain-specific BI applications, creating new possibilities for organizations to build internal generative search systems that leverage their proprietary knowledge.

As a whole, the BI and predictive analytics literature provides the conceptual foundation for the Score stage. Vollrath and Villegas [8] provide the template for reconnecting measurement with strategic objectives, while the application of RAG to domain-specific BI contexts [82,83] demonstrates that the technical infrastructure for capturing generative-search signals is already emerging. The convergence of BI dashboards with RAG-based retrieval creates a dual-purpose infrastructure: the same systems that serve users internally generate the citation and visibility data that the Score stage requires.

Taken together, the eight thematic categories form a coherent landscape that maps onto the 4S lifecycle framework. The SEO evolution and GEO literature (Section 4.1) defines the Surface and Score stages; the RAG and IR literature (Section 4.2) underpins the Signal and Structure stages; knowledge graph research (Section 4.3) provides the architectural foundation for the Structure stage; the digital marketing literature (Section 4.4) informs the Signal and Score stages; conversational AI (Section 4.5) directly shapes the Signal stage; bias and ethics research (Section 4.6) governs all four stages but is most critical for Score; sentiment analysis (Section 4.7) enriches the Signal stage with emotional context; and the BI literature (Section 4.8) provides the measurement infrastructure for the Score stage. No single category addresses all four stages, but together they constitute the complete lifecycle—from capturing conversational intent (Signal) through architecting entity-based data (Structure) and optimizing for LLM citation (Surface) to measuring

visibility in zero-click environments (Score). This convergence is what the Discussion (Section 5.1) synthesizes into trends, challenges, and opportunities.

5. Discussion and Future Research Agenda

5.1. Prevailing Trends

The integrative review reveals several prevailing trends that are reshaping the SEO and BI landscape. First, the transition to AI-generated answer panels is reducing organic click-through rates, challenging the economic model of traditional SEO [6,7]. Organizations must adapt to an environment where visibility does not guarantee traffic, and where “Share of Model” (SoM)—the frequency with which an AI system cites a brand—becomes a primary visibility metric [6]. Second, generative AI dramatically accelerates content production, with studies reporting up to 98.8% efficiency improvements [27]; however, this velocity raises quality and authenticity concerns, as AI content detection tools struggle to accurately identify machine-generated text. Third, the literature reveals a convergence between SEO practices and knowledge engineering, as Schema.org markup, knowledge graphs, and linked data become essential for LLM retrieval [9–11]. This convergence requires SEO practitioners to develop new competencies in structured data and entity-based knowledge representation. Fourth, retrieval-augmented generation is emerging as the dominant architecture for generative search systems [12,13,15]. The rapid advancement of RAG techniques—including dynamic retrieval [40], adaptive retrieval [41], and blended retrieval [38]—suggests that the technical foundations of generative search are still maturing, with significant implications for how content must be structured for retrieval.

5.2. Core Organizational Challenges

The review also identifies several core organizational challenges. The zero-click environment creates attribution gaps that legacy BI frameworks cannot address [8]; organizations lack visibility into when and how generative search engines cite their content, making it difficult to measure the ROI of SEO and GEO investments. Compounding this challenge, generative search systems operate as black boxes, with ranking and citation mechanisms that are opaque to content creators [70,71]. This opacity makes it challenging to optimize content for generative retrieval and raises concerns about fairness and accountability in information access. Explainable AI (XAI) approaches—which aim to render model decision processes transparent and interpretable—offer a promising pathway, yet their application to generative search systems remains largely unexplored. Reliability remains a further concern: LLM-based search systems are prone to hallucinations—generating plausible but incorrect information [36,37]. RAG mitigates but does not eliminate this risk, and organizations must develop strategies for ensuring their content is accurately represented in generative search results. Finally, the use of AI in digital marketing raises significant privacy concerns [74], with tensions between personalization and consumer privacy creating what Saura et al. describe as a “data privacy paradox.” Organizations must navigate GDPR, cookieless tracking, and evolving regulatory frameworks while maintaining effective search and marketing strategies.

5.3. Emerging Strategic Opportunities

Despite these challenges, the review identifies several emerging strategic opportunities. The conversational nature of generative search enables richer intent signals than traditional keyword data [23,24]; organizations that develop capabilities for capturing and analyzing conversational query patterns can gain earlier and deeper insights into customer needs. As generative search engines cite sources, organizations can audit which competitors are being cited for key queries, creating a new form of competitive intelligence [6].

This “LLM audit trail” provides insights into competitor content strategies and authority signals that generative engines prioritize. The shift toward knowledge graph-based search creates opportunities for organizations to establish entity authority—becoming the definitive source for specific entities that LLMs reference [11,26]. This requires investment in structured data, Schema.org markup, and knowledge graph integration. Finally, research on combining RAG with long-context LLMs [42] suggests that hybrid approaches can deliver superior search and retrieval performance, offering organizations a path to more reliable and comprehensive generative search integration. The emerging-technology infrastructure supporting these advances—cloud computing, edge computing, and big-data analytics—will be essential for scaling NLP capabilities to the real-time, conversational demands of generative search [35].

A central contribution of this review is the identification of interdisciplinary intersections that no single field has fully explored. Three bridging intersections are particularly consequential. RAG + BI: Retrieval-augmented systems are not merely technical infrastructure; they generate structured retrieval and citation data that can feed directly into BI dashboards. When a RAG-powered search system retrieves and cites content, it produces an audit trail—which sources were selected, in what order, with what confidence—that organizations can capture, aggregate, and analyze alongside traditional web analytics [82]. Enterprise RAG deployments (Section 4.2) thus become dual-purpose systems: they serve users internally while simultaneously generating the visibility and citation data that the “Score” stage of the 4S framework requires. KG + GEO: Knowledge graphs provide the entity-based data substrate that generative engine optimization depends on. Gonzalez-Garcia et al. [11] demonstrate that Schema.org markup produced for SEO purposes enriches knowledge graphs, while Ibrahim et al. [26] show that LLMs can augment KG construction—creating a virtuous cycle where GEO investment in structured data improves KG quality, which in turn improves generative search visibility. This intersection means that GEO is not purely a content-optimization discipline; it is a knowledge-engineering discipline that requires collaboration between SEO practitioners, data architects, and BI analysts. GEO + BI: The zero-click metrics proposed in the “Score” stage—Share of Model, citation density, brand mention frequency—are inherently BI constructs: they require sampling strategies, audit cadences, competitive benchmarking, and dashboard integration that existing SEO tools were not designed to provide [6,8]. The intersection of GEO and BI creates a new analytical domain where marketing analytics teams must partner with data-engineering teams to build pipelines that capture generative-search signals, define KPIs that lack industry standards, and establish governance protocols for auditing AI-generated answers at scale. Together, these three intersections underscore the interdisciplinary value and originality of the 4S framework: it bridges computer science (RAG, KGs), marketing (GEO, digital marketing), and information systems (BI, analytics) in a way that no existing lifecycle model achieves.

5.4. Future Research Agenda

Based on the identified gaps and opportunities, this review proposes the following research agenda:

1. Developing zero-click attribution frameworks: Future research should develop and validate BI frameworks that measure visibility and impact in zero-click environments, moving beyond click-based metrics to citation density, Share of Model, and influence-oriented KPIs [6].
2. Investigating GEO ranking factors: Empirical studies are needed to identify the factors that influence LLM citation and retrieval in generative search, analogous to the ranking factor studies that shaped traditional SEO [9,30].

3. Addressing algorithmic transparency: Research should develop methods for auditing generative search systems for bias, fairness, and accuracy, building on the frameworks proposed by Dai et al. [70] and Bernard and Balog [71]. Explainable AI (XAI) techniques—such as attention visualization, citation provenance tracking, and counterfactual explanation—represent a particularly promising direction for bridging the gap between algorithmic opacity and practitioner trust.
4. Exploring RAG optimization for enterprise search: Studies should investigate how organizations can implement RAG-based internal search systems that leverage their proprietary knowledge graphs and structured data [45,82].
5. Measuring AI content quality at scale: As AI-generated content becomes ubiquitous, research must address how to maintain quality, authenticity, and search engine compliance at scale [27,34].
6. Integrating conversational analytics into BI: Future work should explore how conversational search data can be integrated into BI dashboards and decision-making processes, bridging the gap between search analytics and strategic planning [8,23].

6. Conclusions

This integrative literature review has synthesized 70 high-value studies from an initial corpus of 11,382 papers filtered to 2962 on-topic studies to map the evolution of SEO in the era of generative AI from a business intelligence perspective. The review makes three primary contributions.

First, it introduces the Signal–Structure–Surface–Score (4S) lifecycle framework, which provides a structured approach to understanding how organizations must adapt their SEO and BI practices across four stages: capturing conversational search intents (Signal), architecting entity-based data structures (Structure), optimizing content for LLM retrieval and GEO (Surface), and defining zero-click attribution metrics (Score). The framework bridges the gap between AI search mechanics and strategic BI measurement, offering practitioners a roadmap for navigating the generative search transition.

Second, the review systematically maps the technical and strategic landscape across eight thematic categories, revealing prevailing trends (zero-click normalization, AI content velocity, SEO–knowledge engineering convergence, RAG as search infrastructure), core organizational challenges (attribution loss, algorithmic opacity, hallucination risks, privacy tensions), and emerging strategic opportunities (real-time intent mapping, competitor LLM audit trails, entity-based content architecture, hybrid RAG–long-context strategies).

Third, it provides a robust future research agenda addressing six priority areas: zero-click attribution frameworks, GEO ranking factors, algorithmic transparency, enterprise RAG optimization, AI content quality measurement, and conversational analytics integration.

The findings demonstrate that the transition from traditional SEO to generative engine optimization is not merely a technical evolution but a fundamental paradigm shift that requires new BI frameworks, new metrics, and new organizational competencies. As LLMs increasingly mediate information access, the ability to structure content for machine readability, optimize for LLM citation, and measure visibility in zero-click environments will become critical organizational capabilities. The 4S framework provides a foundation for developing these capabilities, while the research agenda identifies the empirical and theoretical work needed to advance the field.

The review is not without limitations. The integrative methodology, while enabling broad synthesis, does not provide the systematic rigor of PRISMA-based reviews. The reliance on Google Scholar as a single data source, while justified by the breadth and timeliness required for an integrative review of an emerging field, may introduce coverage biases, particularly for very recent preprints and conference papers, and lacks the strict

quality control of curated databases such as Scopus or Web of Science. Future work should complement these findings with domain-specific systematic reviews that cross-verify the corpus using curated databases (e.g., Scopus, IEEE Xplore, Web of Science, ACM Digital Library) and empirical studies that test the propositions advanced in this paper.

Author Contributions: Conceptualization, K.I.R., D.M., D.S. and C.V.; methodology, K.I.R., D.M., D.S. and C.V.; software, K.I.R. and D.M.; validation, K.I.R., D.M., D.S. and C.V.; formal analysis, K.I.R., D.M., D.S. and C.V.; investigation, K.I.R., D.M., D.S. and C.V.; resources, K.I.R., D.M., D.S. and C.V.; data curation, K.I.R., D.M., D.S. and C.V.; writing—original draft preparation, K.I.R. and D.M.; writing—review and editing, K.I.R., D.M., D.S. and C.V.; visualization, K.I.R. and D.M.; supervision, C.V.; project administration, K.I.R. and D.M.; funding acquisition, K.I.R., D.M., D.S. and C.V. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The bibliographic data used in this study are available on GitHub [21].

Acknowledgments: During the preparation of this work the authors used ChatGPT (gpt-5) (OpenAI) and Grammarly in order to enhance grammatical accuracy and refine sentence structure. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Conflicts of Interest: The authors declare no conflicts of interest.

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